Report 3: Logistic Regression

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1 Implementation

Logistic Regression was implemented from scratch using Gradient Descent.

Prediction model:

$$\hat{y} = \sigma(w_1 x_1 + w_2 x_2 + w_0)$$

Sigmoid function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Binary cross entropy loss:

$$J = -\frac{1}{N} \sum_{i=1}^{N} (y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i))$$

Gradient descent update rules:

$$w_0 = w_0 - r \frac{\partial J}{\partial w_0}$$

$$w_1 = w_1 - r \frac{\partial J}{\partial w_1}$$

$$w_2 = w_2 - r \frac{\partial J}{\partial w_2}$$

The program:

- Reads data from a CSV file
- Computes sigmoid prediction
- Computes binary cross entropy loss
- Updates w_0, w_1, w_2
- Prints intermediate iterative steps

2 Result

Final parameters:

$$w_0 = -18.602$$

$$w_1 = 5.809$$

$$w_2 = 1.797$$

Final loss:

$$Loss = 0.1983$$

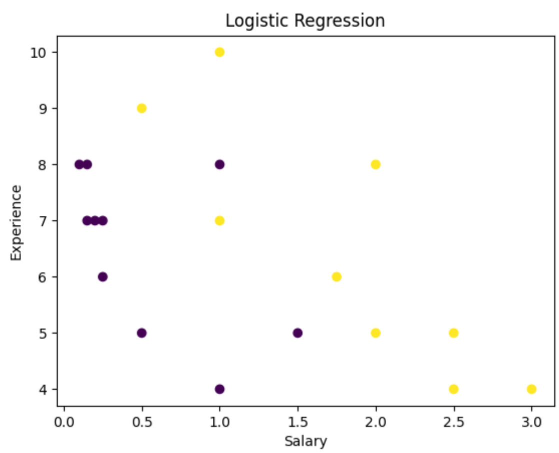

Figure 1: Logistic Regression Visualization

3 Learning Rate Analysis

Different learning rates were tested.

- Large learning rate (1.0): unstable convergence
- Small learning rate (0.001): very slow convergence
- Learning rate 0.1: stable convergence